Lasers in Medical Science (2026) 41:32
https://doi.org/10.1007/s10103-026-04803-w

REVIEW

Light-emitting diode therapy for the prevention and treatment of radiodermatitis in women with breast cancer: systematic review

Fahimeh Nouri[1,2] · Masoumeh Sadeghi[3] · Samira Ebrahimzadeh Zagami[4,5]

Received: 17 September 2025 / Accepted: 5 January 2026
© The Author(s) 2026

Abstract

Breast cancer, the most common cancer worldwide, is treated with surgery, chemotherapy, and radiotherapy. Radiotherapy reduces recurrence and mortality but causes radiodermatitis in over 90% of patients. Current preventive and topical treatments have limited efficacy. Photobiomodulation therapy shows promise for preventing and treating radiodermatitis, but the evidence is limited and controversial. This article aims to systematically review the efficacy of LED therapy in preventing and treating radiodermatitis in women with breast cancer undergoing radiotherapy. This systematic review adheres to PRISMA guidelines, registered in PROSPERO (CRD42023467705), and searches databases (PubMed, Scopus, Cochrane, etc.) for randomized controlled trials (RCTs) and non-randomized trials from 2000 to July 2025. The inclusion criteria targeted human trials of LED therapy for radiodermatitis in breast cancer patients. Study quality was assessed using RoB 2 and ROBINS-I tools. From 882 articles, six studies (three RCTs, three non-randomized; 233 participants) were included. Four studies demonstrated that LED therapy reduced radiodermatitis severity (e.g., 94.7% Grade 0–1 in the LED group vs. 14.3% in the control group; $p<0.0001$) and treatment interruptions. Three studies reported faster healing and pain reduction in acute radiodermatitis, but chronic radiodermatitis showed no improvement. One study found no significant benefit ($p>0.05$). Study quality varied, with two RCTs rated low risk and one high risk. LED therapy shows promise for managing acute radiodermatitis but requires large-scale RCTs with standardized protocols to confirm efficacy, optimize parameters, assess chronic radiodermatitis, and evaluate cost-effectiveness for integration into clinical practice.

Keywords Photobiomodulation · Light-emitted diode · Cancer · Breast cancer · Systematic review · Radiodermatitis

✉ Samira Ebrahimzadeh Zagami
EbrahimzadehZS@mums.ac.ir

Fahimeh Nouri
yas.raha1373@yahoo.com

Masoumeh Sadeghi
sadeghi.masoume@gmail.com

[1] Student Research Committee, Faculty of Nursing and Midwifery, Mashhad University of Medical Sciences, Mashhad, Iran

[2] Instructor of Midwifery, Department of Midwifery, School of Nursing and Midwifery, Birjand University of Medical Sciences, Birjand, Iran

[3] Assisstant Professor of Epidemiology, Department of Epidemiology, School of Health, Mashhad University of Medical Science, Mashhad, Iran

[4] Nursing and Midwifery Care Research Center, Mashhad University of Medical Sciences, Mashhad, Iran

[5] Associated Professor, Faculty of Midwifery, School of Nursing and Midwifery, Mashhad University of Medical Sciences, Mashhad, Iran

Abbreviations

ARD	Acute Radiodermatitis
CRD	Chronic Radiodermatitis
DNA	Deoxyribonucleic Acid
HR	Hazard Ratio
LED	Light-Emitting Diode
LLLT	Low-Level Laser Therapy
NCI	National Cancer Institute
PBMT	Photobiomodulation Therapy
PICO	Population, Intervention, Comparison, Outcome
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROSPERO	Prospective International Registry of Systematic Reviews
RCT	Randomized Controlled Trial
RD	Radiation Dermatitis
RoB 2	Risk of Bias 2 (Cochrane Risk of Bias Tool for Randomized Trials)

Published online: 18 February 2026

ROBINS-I	Risk Of Bias In Non-randomized Studies of Interventions
RT	Radiation Therapy
RTOG	Radiation Therapy Oncology Group
VAS	Visual Analog Scale

Introduction

Breast cancer is the most common cancer diagnosed worldwide, including low- and middle-income countries [1]. The highest incidence rates are observed in North America, Australia, and Western and Northern Europe, while the lowest rates are found in Asia and sub-Saharan Africa [2]. For women with newly diagnosed non-metastatic breast cancer, surgery, radiation therapy, and chemotherapy are all essential components of the multidisciplinary treatment plan. Adjuvant radiation therapy (RT) is typically administered after mastectomy or breast-conserving surgery to eliminate any remaining portions of the tumor. This approach significantly decreases the risk of breast cancer-related death and substantially lowers the chances of recurrence [3]. Radiation therapy works by indiscriminately damaging the DNA of both malignant and normal cells using an ionizing beam. The DNA damage inhibits the cell repair process, triggering apoptosis. The extent of tissue injury and subsequent radiodermatitis depends on the degree of cell destruction [4, 5].

Radiation dermatitis (radiodermatitis) is one of the most common side effects of radiotherapy for cancer. More than 90% of patients undergoing radiotherapy experience some degree of skin reaction [4, 6, 7]. Radiation therapy side effects can be categorized into two groups: early and acute reactions, which happen within 90 days of the treatment beginning, and late consequences, which frequently show up months to years after the radiation treatment is over. High doses of radiation to the skin are the leading cause of this issue in cancer patients. While most patients experience mild or moderate skin reactions, 20–45% develop more severe dermatitis [8, 9]. Radiodermatitis can lead to significant side effects, such as reduced aesthetic appeal, treatment delays, dosage reduction or cessation, and a marked decrease in quality of life [10]. Daily preventive measures and topical or systemic medications are typically prescribed to manage and treat radiodermatitis. Preventive measures include wearing loose clothing, avoiding skin irritation, using mild soap and warm water for washing, keeping the treated area dry and clean, and preventing abrasions, among others [11, 12]. Topical treatments such as glucocorticoids, olive oil, Vaseline, sulfadiazine, and ascorbic acid are also recommended [12–14].

Some systematic reviews have shown that daily cleansing can reduce itching, redness, and scaling, although it does not significantly reduce the overall risk of developing. Regular use of topical corticosteroids has been shown to reduce the incidence of severe dermatitis during radiation therapy, and symptoms can improve within a few weeks [15–17]. However, there is no evidence to support the use of glucocorticoids during radiation therapy to prevent or treat radiodermatitis; their use does not appear to reduce the incidence of this side effect [16]. Photobiomodulation therapy (PBMT) has shown potential in preventing and treating radiation dermatitis (RD). PBMT stimulates or inhibits cellular functions through light exposure at specific wavelengths, intensities, powers, and doses. Photobiomodulation involves non-ionizing radiation sources such as low-level laser therapy (LLLT) and light-emitting diodes (LEDs) [18]. The photons emitted by these sources carry energy absorbed by the mitochondria's cytochromes, which convert it into cellular energy. This process accelerates protein synthesis and cell division, and provides anti-inflammatory and pain-relieving effects that can help restore the skin's texture more rapidly. In vitro studies demonstrated that LED therapy accelerates wound healing, increases procollagen synthesis, reduces inflammatory mediators, and reduces the expression of dermal matrix metalloproteinase (collagenase) in fibroblasts exposed to acute UV damage [19–21].

Despite the potential of PBMT, its application in treating radiodermatitis remains controversial. Some small clinical trials have reported that PBMT with LED reduces the severity of acute dermatitis in breast cancer patients and accelerates skin recovery [22]. Conversely, other studies have presented conflicting findings, indicating limited efficacy of LED or suggesting that its effectiveness depends on specific treatment protocols, such as wavelength, dose, and timing [23]. A systematic review found that LLLT can reduce the severity of oral mucositis in patients undergoing chemotherapy or radiotherapy, but evidence for radiodermatitis was less conclusive, highlighting the need for further research [24]. Another review confirmed LLLT's efficacy in skin wound healing but noted significant variations in treatment parameters and clinical outcomes [25].

In comparison, systematic reviews on LED therapy are fewer and primarily focus on non-cancer-related applications, such as chronic wound healing, acne treatment, or UV-induced skin damage [26]. For example, one systematic review demonstrated that LED therapy has positive effects on diabetic wound healing, though optimizing treatment parameters remains a challenge [27]. However, these reviews do not specifically address radiodermatitis in breast cancer patients and provide limited information on LED's efficacy in this context. The main differences between LLLT and LED include accessibility, cost, and ease of use; LED is considered a more attractive option for broader clinical applications due to its lower cost and suitability for portable

devices [28]. Nevertheless, the absence of a comprehensive review specifically focusing on the use of LED in the prevention and treatment of radiodermatitis in breast cancer patients remains a significant research gap.

Therefore, this study aims to conduct a systematic review of all available research that utilizes phototherapy with Light-Emitting Diodes for the prevention and treatment of radiodermatitis in breast cancer patients. Unlike previous reviews that mainly focused on LLLT or non-cancer-related applications of PBMT, this study will exclusively concentrate on LED to gather existing evidence and guide future research and clinical use.

Methods

The present systematic review was conducted on clinical trial (RCT) studies, and non-randomized clinical trial studies performed based on the PRISMA checklist [29]. The protocol of this systematic review had been registered in the Prospective International Registry of Systematic Reviews (PROSPERO) with the number CRD42023467705 to prevent duplicate publication of systematic reviews.

In this study, the PICO criteria were used to determine the research question, which included: (1) Participants: women with breast cancer undergoing radiotherapy with or without radiodermatitis, (2) Intervention: LED phototherapy, (3) Comparison: LED phototherapy group with control group, (4) Outcomes: Treatment and prevention of radiodermatitis.

Data sources and search strategy

All clinical trials in the field of LED photobiomodulation for radiodermatitis, as LED phototherapy was not introduced before 2000, were searched from 2000 to 2025. The search was conducted in PubMed, Scopus, Cochrane Library, Embase, Google Scholar, Trip database, Web of Science, and Proquest Dissertations & Theses databases without language restrictions, with Boolean operators AND, OR, and other operators based on English keywords: light-emitting diode, radiodermatitis, radiation-induced dermatitis, and breast cancer.

After the initial review of the studies by the first researcher and the elimination of duplicate studies, the titles and abstracts of the remaining studies were reviewed for their inclusion criteria. The references of studies meeting the inclusion criteria were manually examined to identify relevant studies. The steps of searching and selecting studies were recorded according to the PRISMA 2020 flowchart. The second researcher rechecked this path.

Inclusion criteria

Human clinical trials and non-randomized clinical trials, articles, or thesis results related to the use of LED light therapy in the healing and repair of radiodermatitis wounds that met the PICO criteria were considered input studies for review.

Exclusion criteria

Case-control studies, case reports, case series, animal studies, cohort studies, laboratory studies, letters to the editor, editorials, reviews, commentaries, conference abstracts, and studies with irrelevant or incomplete results on the treatment and prevention of radiodermatitis were excluded from the study.

Assessment of study quality

After meeting the inclusion criteria, the full texts of the studies and the dissertation were reviewed and quality assessed using the Cochrane Risk of Bias Tool for Randomized Trials (RoB2). For data extraction and quality assessment of the studies, two researchers (FN, MS) independently performed these tasks. In the event of disagreement, a third researcher (SEZ) provided the final opinion.

Main research tools and their validity and reliability

The Risk of Bias 2 (RoB 2) tool, developed by Cochrane in 2019, evaluates bias in randomized controlled trials (RCTs) across five domains: randomization process, deviations from interventions, missing outcome data, outcome measurement, and selective reporting, with ratings of low risk, some concerns, or high risk (visualized as green, yellow, or red) [30]. The ROBINS-I tool assesses bias in non-randomized studies through seven domains: confounding, participant selection, intervention classification, deviations, missing data, outcome measurement, and selective reporting, rated as low, moderate, severe, or critical [31].

Data extraction

Methodological and required information extracted from studies includes article title, first author, year of publication, country in which the study was conducted, study population, sample size, intervention and control groups, time and days of radiotherapy, receipt of adjuvant treatment, wavelength and time and amount of LED phototherapy, method of dermatitis measurement, improvement measurement tool, outcomes, and article quality score.

Result

A systematic literature search was conducted from 2000 to July 05, 2025, covering various databases. The search yielded 882 articles, of which six studies met the inclusion criteria for this review. The study selection process is summarized in the PRISMA 2020 flowchart (Fig. 1).

The six included studies were conducted in six countries: the United States [32, 33], Germany [34], Morocco [35], China [36], and Brazil [37]. The total sample size across all studies was 233 women with breast cancer undergoing radiotherapy, with individual study sample sizes ranging from 20 to 70 participants. Study designs included three RCTs (two sham-controlled) and three non-randomized trials. Outcomes were assessed using the National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events, Radiation Therapy Oncology Group (RTOG) scale, Visual Analog Scale (VAS), clinical examinations, photography, histology, and questionnaires. Detailed characteristics are presented in Table 1.

The risk of bias assessments for the included studies, as shown in Tables 2 and 3, reveal considerable heterogeneity in methodological quality, affecting the reliability and applicability of the findings. Table 2, which uses the RoB 2 tool for RCTs, shows that one study has a low risk of bias across all areas, suggesting strong randomization, adherence to the protocol, and transparent outcome reporting. In contrast, two RCTs raised concerns or had a high risk, primarily due to issues with randomization and outcome measurement, which could lead to systematic errors that might exaggerate or conceal accurate effect sizes. Table 3, which is based on the ROBINS-I tool for non-randomized studies, shows one study as low risk, indicating minimal confounding and selection biases due to a rigorous design. Conversely,

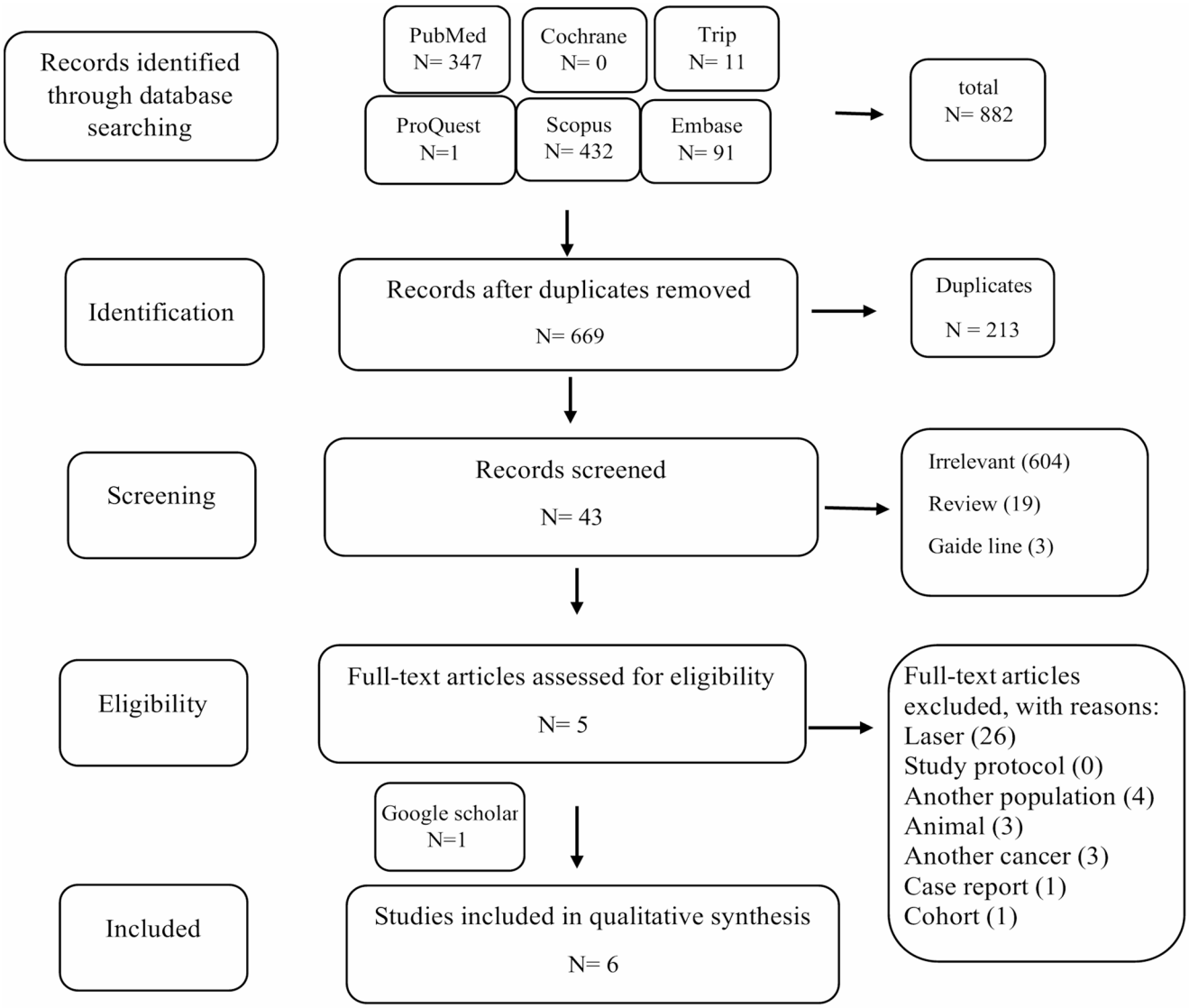


Fig. 1 PRISMA flowchart of the study selection process

Table 1 Systematic review: photobiomodulation for radiodermatitis in breast cancer

Study (Year, Country)	Design and population (Sample Size)	Intervention (PBM Details)	RT Regimen (main field)	Radiotherapy technique & skin-dose modifiers	Outcome measures	Key results (RD Severity, Effect Size)
DeLand (2007, America) [32]	Non-Randomized Breast cancer + RT ($n=47$)	590 nm, 0.15 J/cm², 25 s/session, 250 ms/pulse, 100 pulses, post-RT	≈50–50.4 Gy + 12.6–18 Gy electron boost, 28 fx + 5–10 boost	IMRT + electron boost; bolus NR	NCI + Photograph	LED: 94.7% G0-1, 5.3% G2; Control: 14.3% G1, 85.7% G2-3 ($p<0.0001$). Interruptions: 5.3% vs. 67.9%.
Fife (2010, California) [33]	RCT/blind (sham) Breast cancer + RT ($n=33$)	590 nm, 0.15 J/cm², 35 s/session, 250 ms/pulse, 100 pulses, pre- and post-RT	45–50.4 Gy + boost to 60.4–61.2 Gy, 25–28 + 5–7 fx	3D-CRT, photon NR; bolus only in mastectomy pts	NCI + Questionnaire	LED: 33.3% G1, 66.7% G2; Sham: 6.6% G0, 26.7% G1, 60.0% G2, 6.7% G3 ($p>0.05$). Interruptions: 11.1% (LED) vs. 6.7% (sham).
Strouthos (2017, Germany) [34]	Non-Randomized Breast cancer + RT ($n=70$)	660 + 850 nm, 0.15 J/cm², 60 s/site, 44.6 mW/cm², twice weekly pre-RT	50.4 Gy (1.8 Gy/fraction), 28 fractions	3D-CRT (tangential), photon 6–15 MV; no bolus	NCI + VAS	PBM: 88% G1, 12% G2; Control: 55.6% G1, 40% G2, 4.4% G3; $P=0.0211$; Pain lower in PBM ($p=0.0030$). Interruptions: 0% vs. 4.4%
Hali(2023,Morocco) [35]	Non-Randomized Cancer + RT (15 breast cancer)	660 + 850 nm, 36 J/cm², 8 min/session, 21.6 kJ total energy, twice weekly	40–70 Gy, 15–40 fx (highly variable)	Technique, energy, bolus: all NR	NCI + Clinical exam + Photography	ARD: G1 improved in 2–5 sessions, G2 in 4–8, G3 healed in 8. CRD: no improvement.
Dan (2023, China) [36]	Breast cancer + RT ($n=20$)	650 nm, 30 mW/cm², 10 min/session, every 2 days, 3x/week for 3 weeks	220 cGy/fraction, 25 fractions	Conventional fixed SSD electron + photon supra; no bolus	RTOG + Histology	PBM: 60% G1.5, 10% G2, 20% G2.5, 10% G3; Control: 20% G3, 40% G3.5, 20% G4, 10% G4.5, 10% G5; $P<0.05$
Fischer-Sgrott (2022, Brazil) [37]	RCT/blind (sham) Breast cancer + RT ($n=48$)	660 nm, 3 J/point, 10 min/session, every 2 cm across breast	Median 50.4 Gy (42–55 Gy), mostly 25–28 fx, 1.8–2 Gy/fx	3D-CRT, photon (mostly 6 MV); bolus only in mastectomy pts	RTOG + VAS +	Breast plastron radiodermatitis 11.5% vs. 59.1% (HR 0.090, $p=0.001$); non-breast area 88.5% vs. 86.4% (HR 1.21, $p=0.82$)

Table 2 Risk of bias assessment (RoB 2) for RCTs

Study	Randomization process	Deviations from intervention	Missing outcome data	Measurement of outcome	Selection of reported result	Overall
Fife (2010)	Some concern	Low	Low	Low	Low	Some concern
Fischer-Sgrott (2022)	Low	Low	Low	Low	Low	Low
Li Dan (2023)	Some concern	Some concern	Low	Some concern	Some concern	High

Table 3 Risk of bias assessment (ROBINS-I) for cohort study and non-randomized trials

Study	Confounding	Selection of participants	Classification of interventions	Deviations from interventions	Missing outcome data	Measurement of outcome	Selection of reported result	Overall
Hali (2023)	High	Low	low	Low	Low	Moderate	low	High
DeLand (2007)	Moderate	Moderate	Low	Low	Low	Low	Low	Moderate
Strouthos (2017)	low	Low	Low	Low	Low	Low	Low	Low

another study was rated as moderate risk due to issues with participant selection and confounding. At the same time, a third was deemed high risk, primarily due to significant confounding and selection biases that likely affected the validity of the results.

Effects of LED phototherapy on radiodermatitis

Prevention and severity reduction

Four studies [32, 34, 36, 37] evaluated LED phototherapy for the prevention of radiodermatitis, focusing on reducing its incidence or severity (treatment initiated before or concomitant with radiotherapy and, in one study, extended post-radiotherapy). DeLand (2007) reported 94.7% of the LED group had Grade 0–1 radiodermatitis compared to 14.3% in the control group ($p < 0.0001$), with fewer treatment interruptions (5.3% vs. 67.9%) [32]. Strouthos (2017) found 88% Grade 1 and 12% Grade 2 in the LED group compared to 55.6% Grade 1, 40% Grade 2, and 4.4% Grade 3 in the control group ($p = 0.0211$), with lower pain scores ($p = 0.0030$) and no interruptions [34]. Dan (2023) observed milder radiodermatitis in the LED group (60% Grade 1.5, 10% Grade 2) compared to the control group (20% Grade 3, 40% Grade 3.5, $p < 0.05$) [36]. Fischer-Sgrott (2022) reported an 11.5% incidence of breast plastron radiodermatitis in the LED group compared to 59.1% in the sham group (HR 0.090, $p = 0.001$) [37] (Table 1).

Combined prevention and treatment

Fife (2010) examined both prevention and treatment, found no significant difference between LED and sham groups ($p > 0.05$), with similar interruption rates (11.1% vs. 6.7%) [33] (Table 1).

RT radiation therapy, *NCI* National Cancer Institute scale, *RTOG* Radiation Therapy Oncology Group scale, *ARD* acute radiodermatitis, *G* grade, *IMRT* intensity-modulated radiotherapy, *D-CRT* three-dimensional conformal radiotherapy, *fx* fraction(s), *NR* not reported.

Discussion

This systematic review is the first to evaluate LED phototherapy for preventing and managing radiodermatitis in women with breast cancer undergoing radiotherapy, addressing a significant gap in the literature. Synthesizing six studies with 233 participants across diverse regions, this analysis highlights the potential of LED therapy, emphasizing its affordability, ease of use, and suitability for portable devices. Unlike prior reviews that focus on photobiomodulation therapy for radiodermatitis, this study uniquely targets LED therapy for its treatment.

Fischer-Sgrott [37] and Strouthos [34] (high-quality studies) demonstrate that LED phototherapy reduces radiodermatitis severity, pain, and treatment interruptions compared to controls, due to photobiomodulation's enhancement of mitochondrial function, decreasing inflammation, and promoting tissue healing [21, 38]. Consistent results across wavelengths (660–850 nm) suggest versatility, though efficacy for chronic radiodermatitis remains unproven, warranting cautious interpretation [38]. The limited number of low-risk studies prevents the endorsement of LED therapy as a standard of care.

In contrast, the four remaining studies reported inconsistent or inconclusive results [32, 33, 35, 36]. A detailed examination of these discordant findings reveals that the discrepancies are largely attributable to two principal factors. The most notable is the extreme dosimetric deviation observed in Hali (2023), which is ten times higher than the 3–6 J/cm² used in low-risk studies. This excessive energy density clearly contradicts the well-established biphasic dose–response relationship (Arndt–Schulz law) that regulates photobiomodulation, leading to a transition from biostimulatory to bioinhibitory effects on mitochondrial respiration and anti-inflammatory pathways [39–41].

The second factor is methodological weakness: the absence of a concurrent control group in Hali (2023), the use of historical controls in DeLand (2007), inadequate blinding and outcome assessment in Fife (2010), and potential selection and confounding bias from consecutive allocation in Dan (2023). When these factors are missing, the results are consistently positive and robust. Thus, the apparent contradictions in the literature are resolved completely when evidence is stratified simultaneously by methodological rigour and the biological plausibility of the applied parameters.

Previous reviews, such as Gobbo (2023) [42], Rocha (2022) [22], Aguiar (2021) [43], and Lin (2025) [44], demonstrate photobiomodulation's efficacy for radiodermatitis, but focus solely on LLLT; none of these studies include LED research. These reviews showed inconsistent LLLT results for radiodermatitis due to protocol variability, whereas our low-risk studies provide more consistent evidence for LED therapy. LED's advantages over LLLT include lower cost, broader beam coverage for larger treatment areas, simpler administration requiring less specialized training, and suitability for portable, home-use devices, enhancing patient accessibility [45].

LED phototherapy shows promise for treating acute radiodermatitis, but its routine use requires validation. Clinicians may consider it with portable devices and low-risk protocols, pending the establishment of standardized

parameters. Future research should include large-scale, low-risk RCTs to confirm effectiveness, optimize settings, and compare with LLLT and corticosteroids. Studies on chronic radiodermatitis over time, as well as cost-effectiveness analyses, are also essential. In conclusion, high-quality evidence suggests LED therapy has potential, but further rigorous research is necessary to establish it as a standard of care, building on the insights of this pioneering review.

Limitations

The small number of studies ($n=6$) and the few rated as having low risk of bias limit the strength of the conclusions. Heterogeneity in treatment protocols, including variations in wavelength, dose, and frequency, as well as diverse outcome measures (e.g., NCI, RTOG, VAS), precluded meta-analysis, limiting the ability to quantify the effect size of LED therapy. Future research should prioritize large-scale randomized controlled trials (RCTs) with standardized PBMT protocols (e.g., 660 nm, 1–3 J/cm², twice weekly) to validate efficacy across diverse populations and RD stages. Investigations into the long-term effects of PBMT on skin integrity, cosmetic outcomes, and patient-reported quality of life are also warranted. Additionally, cost-effectiveness analyses are crucial for evaluating the feasibility of PBMT in resource-constrained settings, thereby supporting its global adoption.

Conclusion

This systematic review establishes that LED-based PBMT effectively mitigates acute radiodermatitis in breast cancer patients when standardized protocols and rigorous methodologies are used. Studies with consistent PBMT parameters and robust designs demonstrate significant improvements in RD severity, pain, and patient comfort. In contrast, inconsistent outcomes in other studies stem from variable protocols and methodological limitations. To establish LED as a standard of care, future RCTs with harmonized protocols, validated outcome measures, and evaluations of chronic RD and cost-effectiveness are essential. These efforts will facilitate LED's integration into clinical practice, enhancing RD management and improving patient outcomes worldwide.

Acknowledgements The authors sincerely thank the Nursing and Midwifery Care Research Center of Mashhad University of Medical Sciences for their invaluable support and cooperation in facilitating this research. This text has been translated into English and improved in quality using AI and Grammarly.

Author contributions FN conceptualized the study, conducted the literature search, data extraction, quality assessment, and drafted the manuscript. MS performed data extraction, quality assessment, and contributed to writing the manuscript. SEZ supervised the study, reviewed the methodology, and revised the manuscript for critical intellectual content. All authors read and approved the final manuscript.

Funding The authors received no financial support for this work, nor did they obtain any funding for this study.

Data availability All data analyzed in this systematic review are derived from publicly available studies included in the review. The complete list of included studies and their references is provided in the manuscript. No additional datasets were generated or analyzed.

Declarations

Ethics approval and consent to participate Not applicable. This study is a systematic review of existing literature and does not include primary data collection from human participants.

Consent for publication Not applicable. This manuscript does not contain any person's data in any form.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Cancer IAfRo GLOBOCAN 2020: new global cancer data: union for international cancer control (UICC). https://www.uicc.org/news
2. Sung H, Ferlay J, Siegel RL et al (2021) Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: a cancer journal for clinicians 71(3):209–249
3. Group EBCTC (2011) Effect of radiotherapy after breast-conserving surgery on 10-year recurrence and 15-year breast cancer death: meta-analysis of individual patient data for 10 801 women in 17 randomised trials. Lancet 378(9804):1707–1716
4. Bray FN, Simmons BJ, Wolfson AH, Nouri K (2016) Acute and chronic cutaneous reactions to ionizing radiation therapy. Dermatol Ther 6(2):185–206
5. Gianfaldoni S, Gianfaldoni R, Wollina U, Lotti J, Tchernev G, Lotti T (2017) An overview on radiotherapy: from its history to its current applications in dermatology. Open Access Maced J Med Sci Jul 25(4):521–525
6. Leventhal J, Young MR (2017) Radiation dermatitis: recognition, prevention, and management. Oncol (08909091) 31(12)

7. Backler C, Bruce SD, Suarez L, Ginex PK (2020) Radiodermatitis: clinical Summary of the ONS Guidelines™ for Cancer Treatment-Related Radiodermatitis. Clin J Oncol Nurs 24(6):681–684. https://doi.org/10.1188/20.CJON.681-684
8. Bernier J, Bonner J, Vermorken J et al (2008) Consensus guidelines for the management of radiation dermatitis and coexisting acne-like rash in patients receiving radiotherapy plus EGFR inhibitors for the treatment of squamous cell carcinoma of the head and neck. Ann Oncol 19(1):142–149
9. Santa Cruz O, Tsoutsou P, Castella C et al (2018) Locoregional control and toxicity in head and neck carcinoma patients following helical tomotherapy-delivered intensity-modulated radiation therapy compared with 3D-CRT data. Oncology 95(2):61–68
10. de Andra Fuzissaki M, Paiva CE, de Oliveira MA, Canto PPL, de Paiva Maia YC (2019) The impact of radiodermatitis on breast cancer patients' quality of life during radiotherapy: a prospective cohort study. J Pain Symptom Manage 58(1):92–99. e91
11. Yang X, Ren H, Guo X, Hu C, Fu J (2020) Radiation-induced skin injury: pathogenesis, treatment, and management. Aging (Albany NY) Nov 16(22):23379–23393
12. Finkelstein S, Kanee L, Behroozian T et al (2022) Comparison of clinical practice guidelines on radiation dermatitis: a narrative review. Support Care Cancer 30(6):4663–4674
13. Lee J, Lee SW, Hong JP, Shon MW, Ryu SH, Ahn SD (2016) Foam dressing with epidermal growth factor for severe radiation dermatitis in head and neck cancer patients. Int Wound J 13(3):390–393
14. Menon A, Prem SS, Kumari R (2021) Topical betamethasone valerate as a prophylactic agent to prevent acute radiation dermatitis in head and neck malignancies: A Randomized, Open-Label, phase 3 trial. Int J Radiat Oncol Biol Phys Jan 1(1):151–160
15. Hindley A, Zain Z, Wood L et al (2014) Mometasone furoate cream reduces acute radiation dermatitis in patients receiving breast radiation therapy: results of a randomized trial. Int J Radiat Oncol Biol Phys Nov 15(4):748–755
16. Chan RJ, Webster J, Chung B, Marquart L, Ahmed M, Garantziotis S (2014) Prevention and treatment of acute radiation-induced skin reactions: a systematic review and meta-analysis of randomized controlled trials. BMC Cancer 14(1):53
17. Ulff E, Maroti M, Serup J, Nilsson M, Falkmer U (2017) Prophylactic treatment with a potent corticosteroid cream ameliorates radiodermatitis, independent of radiation schedule: a randomized double blinded study. Radiother Oncol 122(1):50–53
18. Zhang WW, Wang XY, Chu YX, Wang YQ (2022) Light-emitting diode phototherapy: pain relief and underlying mechanisms. Lasers Med Sci 2343–2352. https://doi.org/10.1007/s10103-022-03540-0
19. Weiss RA, McDaniel DH, Geronemus RG et al (2005) Clinical experience with light-emitting diode (LED) photomodulation. Dermatol Surg 31:1199–1205
20. Weiss RA, McDaniel DH, Geronemus RG, Weiss MA (2005) Clinical trial of a novel non-thermal LED array for reversal of photoaging: clinical, histologic, and surface profilometric results. Lasers Surg Med 36(2):85–91
21. Camargo CP, Carvalho HA, Maluf FC, Sousa AADC, Perin POM, Perin MM, Morais-Besteiro J, Gemperli R (2021) Light-emitting diodestimulates radiodermatitis recovery. Acta Cir Bras 36(3):e360301. https://doi.org/10.1590/ACB360301
22. Rocha SR, da Costa Ferreira SA, Ramalho A, Santos VLCG, Nogueira PC (2022) Photobiomodulation therapy in the prevention and treatment of radiodermatitis in breast cancer patients: systematic review. J Lasers Med Sci 13:e42
23. Rao D, Dsouza CN, Prabhu SS, Kumar P, Prabhu V (2024) Photobiomodulation therapy for mitigating severity of radiodermatitis in cancer patients undergoing radiotherapy: a scoping review. Support Care Cancer 32(11):1–17
24. Peng J, Shi Y, Wang J et al (2020) Low-level laser therapy in the prevention and treatment of oral mucositis: a systematic review and meta-analysis. Oral Surg, Oral Med, Oral Pathol, Oral Radiol 130(4):387–397. e389
25. Seyyedi SA, Taram S, Heydari M, Valizadeh R (2022) Efficacy of low-level laser therapy in oral mucosal surgical wound healing: a systematic review and meta-analysis. Explor Med 3(5):451–460
26. da Rocha RB, Araújo DD, Machado FDS, Cardoso VS, Araújo AJ, Marinho-Filho JDB (2024) The role of light emitting diode in wound healing: a systematic review of experimental studies. Cell Biochem Funct 42(5):e4086
27. Baracho VdS, Silva NCd, Peixoto MFD, Sampaio KH, Cordeiro CAF, Lucas TC (2023) LED phototherapy in tissue repair of chronic wounds in people with diabetes: a systematic review. Rev Gaúcha Enferm 44:e20220274
28. Jagdeo J, Austin E, Mamalis A, Wong C, Ho D, Siegel DM (2018) Light-emitting diodes in dermatology: a systematic review of randomized controlled trials. Lasers Surg Med 50(6):613–628
29. Page MJ, McKenzie JE, Bossuyt PM et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. Bmj Mar 29:372:n71
30. Sterne JAC, Savović J, PageMJ, Elbers RG, Blencowe NS, Boutron I, Cates CJ, Cheng HY, Corbett MS, Eldridge SM, EmbersonJR, Hernán MA, Hopewell S, Hróbjartsson A, Junqueira DR, Jüni P, Kirkham JJ, Lasserson T, Li T, McAleenan A, Reeves BC, Shepperd S, Shrier I, Stewart LA, Tilling K, White IR, Whiting PF, Higgins JPT (2019) RoB 2: a revised tool for assessing risk of bias in randomised trials. BMJ 366:l4898. https://doi.prg/10.1136/bmj.l4898
31. Sterne JA, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, Henry D, Altman DG, Ansari MT, Boutron I, Carpenter JR, Chan AW, Churchill R, Deeks JJ, Hróbjartsson A, Kirkham J, Jüni P, Loke YK, Pigott TD, Ramsay CR, Regidor D, Rothstein HR, Sandhu L, Santaguida PL, Schünemann HJ, Shea B, Shrier I, Tugwell P, Turner L, Valentine JC, Waddington H, Waters E, Wells GA, Whiting PF, Higgins JP (2016) ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ 12(355):i4919. https://doi.org/10.1136/bmj.i4919
32. DeLand MM, Weiss RA, McDaniel DH, Geronemus RG (2007) Treatment of radiation-induced dermatitis with light-emitting diode (LED) photomodulation. Lasers Surg Med 39(2):164–168
33. Fife D, Rayhan DJ, Behnam S et al (2010) A randomized, controlled, double-blind study of light emitting diode photomodulation for the prevention of radiation dermatitis in patients with breast cancer. Dermatol Surg 36(12):1921–1927
34. Strouthos I, Chatzikonstantinou G, Tselis N et al (2017) Photobiomodulation therapy for the management of radiation-induced dermatitis: A single-institution experience of adjuvant radiotherapy in breast cancer patients after breast conserving surgery. Strahlenther Onkol 193(6):491–498
35. Hali F, ElKissouni A, Chiheb S, Charkaoui M, Berreda Y, Benchakroune N, Sahraoui S. Light-emitting diodetherapy for the management of radiodermatitis. Our Dermatology Online/Nasza Dermatol Online 14(2). https://doi.org/10.2203/dose-response.09-027.Hamblin
36. Dan L, Xuewu F, Lei D, Long T, Yimin H (2023) Application of light-emitting diode to prevent acute radiodermatitis in patients with breast cancer. J Radiation Res Radiation Process 40(2):54–59
37. Fischer-Sgrott F, Sapienza L, Rech P et al (2022) Photobiomodulation for radiodermatitis prevention in breast cancer: results from a double blind randomized controlled trial (PHOTODERMIS trial). Int J Radiat Oncol Biol Phys 114(3):e35
38. Camargo CP, Forner-Cordero A, Silva BM et al (2023) Effect of photobiomodulation with different wavelengths on radiodermatitis treatment. Plast Reconstr Surgery–Global Open 11(2):e4809

39. Huang YY, Chen AC, Carroll JD, Hamblin MR (2009) Biphasic doseresponse in low level light therapy. Dose Response 7(4):358–383. https://doi.org/10.2203/dose-response.09-027.Hamblin
40. Hamblin MR (2018) Mechanisms and mitochondrial redox signaling in photobiomodulation. Photochem Photobiol 94(2):199–212
41. Salehpour F, Sadigh-Eteghad S, Mahmoudi J, Kamari F, Cassano P, Hamblin MR (2023) Biphasic Dose–responsein Photobiomodulation of Neuronal Cells and the Brain. In: Photobiomodulation for the Brain. Synthesis Lectures on Biomedical Engineering. Springer, Cham. https://doi.org/10.1007/978-3-031-36231-6_5
42. Gobbo M, Rico V, Marta GN et al (2023) Photobiomodulation therapy for the prevention of acute radiation dermatitis: a systematic review and meta-analysis. Support Care Cancer 31(4):227
43. de Aguiar BRL, Guerra ENS, Normando AGC, Martins CC, Dos Reis PED, Ferreira EB (2021) Effectiveness of photobiomodulation therapy in radiation dermatitis: a systematic review and meta-analysis. Crit Rev Oncol Hematol 162:103349
44. Lin Y-T, Tung KM, Chiou J-F, Chen Y-C, Hou W-H (2025) Effects of photobiomodulation therapy for acute radiation dermatitis in patients with cancer: a systematic review and meta-analysis of real-world evidence. Radiother Oncol 202:110589
45. Fornaini C, Wang H, Li Y, Rocca JP (2024) Light-Emitting-Diode photochemical effects in dentistry: an overview. Laser Ther 31(1). https://doi.org/10.4081/ltj.2024.377

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.